Free Fall: Measuring *g* from a Ticker Tape

SYNOPSIS

Every problem you solved in Module 1 used a number you were simply handed: 9.81 m/s². In this lab you measure it yourself. A **ticker-tape timer** drags a paper tape through a vibrating stylus and stamps a dot on it forty times a second while a mass pulls the tape toward the floor. Those dots are a position-versus-time graph that the falling mass drew for you — you just have to measure it with a ruler. You will fit a curve to that data, read *g* straight out of the fit, and then argue about how close your answer really is.

Due See the Module 2 Lab folder on Brightspace

Pairs with M2 N3

Report Full lab report — use the Lab Template

Data Your own — three tapes per group

Print PDF version of this page

PERFORMANCE EXPECTATIONS

By the end of this lab, you will be able to…

- **PE1** **Read a ticker tape as position-versus-time data** — measure cumulative distance from a chosen origin dot, and convert dot number to time using the timer frequency, $t = n/f$.
- **PE2** **Map a kinematic model onto a fit equation** — recognize that $x = x_0 + v_0\Delta t + \tfrac{1}{2}a(\Delta t)^2$ and $y = Ax^2 + Bx + C$ are the same equation, and read the acceleration out of the fit as $a = 2A$.
- **PE3** **Interpret every fit parameter physically** — say what A, B and C each mean on your tape, and use B to work out where on the fall your dot 0 actually sits.
- **PE4** **Build a graph that can be read without you standing next to it** — correct variable on each axis, units in both axis titles, markers rather than joined dots, the fitted curve drawn over the data, and the fit equation and R^2 displayed on the chart.
- **PE5** **Report a measured value with an uncertainty** — mean and standard deviation across your three tapes, carried to the correct number of significant digits.
- **PE6** **Compute percent error and apply the 10% rule** — and, if your error is large, identify a specific physical mechanism and estimate its size rather than waving at it.

🔎 Why a tape and not a stopwatch?

Drop something from a metre up and it lands in about 0.45 s. Your reaction time is roughly 0.2 s. A stopwatch measurement of that fall is not slightly imprecise — it is dominated by you. Timing is almost always the weakest measurement in a mechanics experiment, and the standard fix is to *stop timing by hand*.

The ticker-tape timer does exactly that. It ticks at a fixed rate set by the AC line frequency, so the time axis of your data is essentially perfect and free: dot number n happened at $t = n/f$, no human involved. That moves the entire measurement onto the one axis you *can* measure well — distance, with a ruler, on a tape lying flat on a table. It also hands you twenty-odd data points instead of one, which is what makes fitting possible at all.



Time

Set by the timer, not by you. $t = n/f$, exact to better than 1%.



Distance

The measurement you actually make. A millimetre ruler, read carefully, is good to ±0.5 mm.



Acceleration

Never measured directly. It comes out of a fit to the other two — which is why the fit has to be the right one.

Predict first — before you run a single tape. The falling mass does not just fall: it also has to *drag a paper tape through the timer*, past a stylus that is striking it forty times a second. If that contact resists the motion at all, what will the acceleration you measure from the tape do?

(A) Come out higher than 9.81 m/s².

(B) Come out lower than 9.81 m/s².

(C) Stay at 9.81 — friction changes the speed, not the acceleration.

(D) Scatter randomly above and below 9.81.

What you need

- Ticker-tape timer. Most of our units tick at **40 Hz** — **check the label on the one you are using**, because some run at 60 Hz and every number you compute depends on it.
- Paper tape, a hanging mass (200–500 g works well), masking tape, and a clear drop of at least 1.5 m — a table edge or the stairwell rail.
- A metre stick or a two-metre stick with millimetre divisions.
- A spreadsheet program — Excel, Google Sheets, or LibreOffice Calc.
- The **Lab Template** and the **Writing Lab Reports** module on Brightspace. Read both before you start writing.

Part I — Running the tape

A ticker-tape timer marks a paper tape at a fixed frequency while the tape is pulled through it. Because the marks are made at equal time intervals, the *spacing* between marks is a direct picture of the speed: evenly spaced dots mean constant speed, and dots that spread out mean the object is speeding up. A usable tape runs about 1.0–1.5 m.

1. Thread the tape through the timer and attach the hanging mass to one end. Check that the tape runs freely and is not twisted. **A twisted tape binds, and binding is the single largest source of error in this method.**
2. Hold the mass so the tape is slack-free but not under tension, with the timer clamped at the table edge and the mass free to fall clear of everything below it.
3. Start the timer *first*, then release the mass. Catch the tape — or let the mass land on something soft.
4. Lay the tape out flat on the bench and tape it down so it cannot shift while you measure.
5. Repeat until you have **three good tapes**. A tape is good if the dots are distinct and separating smoothly; discard and re-run any tape with a visible skip, a smear, or a section where the spacing jumps around.

> **Sign convention here.** The physics is **$a = -g = -9.8\ \text{m/s}^2$**: upward is positive, and $g = 9.8\ \text{m/s}^2$ is a magnitude that never carries a sign. On this tape you measure distances *down the tape as positive*, so what you plot and fit is the *magnitude* of the distance fallen as it grows with time. That is why the fit returns $A \approx +4.9\ \text{m/s}^2$ and B positive rather than negative values. It is a convenience for measuring and fitting, not a change to the physics.

Part II — Measuring the dots

This is the whole experiment. Everything after it is arithmetic.

1. **Find your origin dot.** Near the start of the tape you will find a dark blot or a cluster of dots almost on top of each other — that is the timer striking the same spot while the mass was still held, plus the first few ticks of the fall, which are only a few millimetres apart. Skip past it. The first dot you can clearly separate from its neighbours is your **dot 0**. Circle it and label it 0.
2. **Number the rest.** Circle and number every subsequent dot: 1, 2, 3, … Keep going as far as the dots stay clean, ideally 20 or more.
3. **Measure from dot 0 to each dot.** Not dot-to-dot. Measuring each gap separately and adding them up compounds every small error you make; measuring from a single origin does not. Record to the nearest millimetre, in centimetres, in your lab notebook.
4. Measure to the *dot*, not to the circle you drew around it.
5. Do all three tapes. Keep each one's data in its own column.

> **Your dot 0 is not the release point, and that is fine.** By the time the dots are far enough apart to measure, the mass has already been falling for a tick or two and is moving at a few tenths of a metre per second. This is unavoidable — the timer is running before you let go, the first gaps are smaller than the dots themselves, and you do not release in sync with the

ticks. The analysis below is built so that this costs you nothing. In fact, the fit will tell you *how fast* the mass was already moving at dot 0, and you will use that number in your discussion.

Building your data table

In your spreadsheet, for each tape, build these columns:

Table 1 — required columns, one block per tape

Column	What goes in it
Dot number n	0, 1, 2, 3, …
Time t (s)	$t = n/f$. At 40 Hz, dot 7 is at 7/40 = 0.175 s.
Distance (cm)	What you measured from dot 0, straight from your notebook. Do not "clean up" a value that looks out of line.
Position x (m)	The distance column divided by 100. **Every fit below is done in metres and seconds.**

Units are not a formality here. If you fit centimetres against seconds you will get an acceleration of about 980, and it will look like a plausible number in a cell. Convert to metres *before* you fit, and show the conversion as one of your sample calculations.

Part III — Making the graph and reading the fit

This part is the analysis, and it is where most of the marks are. You will plot position against time, fit a second-order polynomial to it, and read the acceleration out of the fit. Read this section before you open the chart tool, not after.

Why a quadratic, and not something else

You already know the equation that describes this motion. It is the one from Module 1:

$$x = x_0 + v_0\,\Delta t + \tfrac{1}{2}\,a\,(\Delta t)^2$$

position under constant acceleration

And a spreadsheet's order-2 polynomial trendline fits this:

$$y = A\,x^2 + B\,x + C$$

generic parabola — what the trendline reports

These are the same equation written with different letters. That is the whole trick of this lab. Line them up term by term, with your **time** column playing the role of *x* and your **position** column the role of *y*. You are not fitting a parabola because the data happens to look curved; you are fitting one because constant acceleration *is* a quadratic in time, and the coefficients of that quadratic are the physical quantities you came to measure.

What each fit parameter means

Table 2 — reading the three coefficients, with *y* in metres and *x* in seconds

Fit reports	Physics calls it	Units	What you do with it
A	$\frac{1}{2}a$	m/s²	$a = 2A$ — this is your measured acceleration. **Double it. Every time.** On a good tape *A* itself lands near 4.9, not near 9.8.
B	v_0	m/s	The speed the mass already had at dot 0 — a few tenths of a m/s, and positive. **Interpret it in your discussion; do not just report it.**
C	x_0	m	The position of dot 0 in your own coordinate system. It should come out near zero, because you measured everything from dot 0.

Notice that you get all three for free. You are not assuming the mass started from rest, and you are not assuming dot 0 is anywhere in particular — the fit works those out and hands them to you. That is precisely why this method survives a dot 0 that is already moving.

Making the plot, step by step

1. Select your **time** column and your **position-in-metres** column, and insert a **Scatter (XY)** chart. Not a line chart, not a bar chart. Time goes on the horizontal axis — the independent variable always does.
2. Choose the **markers-only** scatter subtype. Do not let the spreadsheet join your points with a line: a line claims you measured the motion in between the dots, and you did not. The only curve on the chart should be the fit.
3. Right-click any data point and choose **Add Trendline**.
4. In the trendline options, select **Polynomial** and set the **Order to 2**. Leave **Set Intercept** unchecked — forcing the curve through zero throws away the *C* you want to read.
5. Check both **Display Equation on chart** and **Display R-squared value on chart**. You may need to scroll down in the options panel to find them.
6. Give the chart **axis titles with units** — "Time (s)" and "Position (m)" — and delete the default chart title and the legend, which say nothing on a single-series plot. The caption below the figure does that job in a report.
7. Read *A*, *B*, *C* and R^2 off the chart. **Double *A*.** That is your measured acceleration for that tape.
8. Repeat for all three tapes. Each tape gets its own graph in your report, with the equation and R^2 visible.

Widen the equation label before you trust it. Excel displays trendline coefficients with very few significant digits by default, and for A that can round your answer by a percent or more. Click the equation label, open **Format Trendline Label**, and set the number format to **Scientific** with 4 decimal places — or at minimum confirm you are reading four significant digits out of A. Rounding A to 4.9 and doubling it is not a measurement; it is the number you were handed in Module 1 given back to you.

What every graph in this report must show

Table 3 — the checklist your figures are graded against

Element	Requirement
Horizontal axis	Time, in seconds. Titled **Time (s)**. The independent variable.
Vertical axis	Position, in metres. Titled **Position (m)**. Never centimetres.
Data	Plotted as **individual markers**, unconnected. One tape per chart.
Fit	The order-2 curve drawn *over* the data, so a reader can see where it fits well and where it does not.
Fit equation	Displayed on the chart, with enough significant digits in A to be read (four).
Goodness of fit	R^2 displayed on the chart.
Scale	Axes covering your data and little else — a plot where the parabola occupies a fifth of the frame hides exactly the curvature you are measuring.
Figure number	Numbered in order of first mention: Figure 1, Figure 2, …
Caption	*Below* the figure, one sentence saying what is plotted and what the curve is.
Mention in text	Introduced by a sentence before it appears — "Figure 2 shows…". A figure nobody refers to earns nothing.
No chart junk	No default "Chart Title", no legend on a one-series plot, no gridline clutter, no 3-D anything.

Sanity-check the three numbers before you write them down. On a good tape, $A \approx 4.9$ m/s², B is a few tenths of a m/s and positive, and C is within a centimetre or so of zero. If A comes back near 490 you are still in centimetres. If A is near 9.8 you have fitted something other than position against time — or you are about to double a number that was already the acceleration. If B is large and negative, your distances are running the wrong way down the tape.

⚠ Why we are not plotting x against t^2

You may have seen free fall handled by plotting position against *time squared*, so that $x = \frac{1}{2}gt^2$ becomes a straight line whose slope is $g/2$. That is a legitimate trick, and it works — **as long as the object starts from rest at the point you call zero.**

On a ticker tape, it does not. Your dot 0 is a tick or two into the fall, so v_0 is not zero, and the real relationship is $x = v_0t + \frac{1}{2}gt^2$. Plotted against t^2, that is *not* a straight line — the v_0t term becomes $v_0\sqrt{(t^2)}$, which curves. Fitting a line to it anyway does not simply shift the intercept; it quietly folds part of the velocity term into the slope and pushes your g **upward**, typically by 10–15% on a real tape.

What makes this worth a paragraph is that the wrong answer looks *better*. Because the tape also drags, which pushes g down, the two errors partly cancel and the linearized fit lands suspiciously close to 9.8 — for the wrong reason, with an R^2 of 0.999 reassuring you the whole time. The order-2 fit avoids the trade entirely: it models v_0 instead of wishing it away.

Analysis

Your report needs a results table that gathers every fit. Build it in the spreadsheet, and number, caption and introduce it in the text ("Table 2 shows…").

Required columns for your fit-results table

Column	What goes in it
Tape	1, 2, 3
Dots used	How many, and which range — part of your data, not a detail
A, B, C	Trendline coefficients, with units
R^2	From the trendline display
$g = 2A$	Measured acceleration, m/s²
% error	Against 9.81 m/s²

The statistics you need

You have three tapes, so you can report a mean and a spread.

$$\bar{g} = (1/N)\,\Sigma\, g_i \qquad s = \sqrt{[\,\Sigma(g_i - \bar{g})^2 \,/\, (N-1)\,]}$$

In Excel: `=AVERAGE(range)` and `=STDEV.S(range)`

Quote your result as $\bar{g} \pm s$, rounded so the uncertainty carries one significant digit and the value is rounded to the same decimal place. A mean of 8.6127 with a standard deviation of 0.3142 is reported as **8.6 ± 0.3 m/s²** — never as 8.6127 ± 0.3142.

> **Three tapes is a small sample, and you should say so.** A standard deviation from $N = 3$ is itself very uncertain. It tells you roughly how much one tape differs from another; it does not license a precise claim about how well you know the average. Use it, quote it, and note the limitation in your discussion rather than pretending it is more than it is.

Percent error

$$\%\ \text{error} = \left| \text{measured} - \text{accepted} \right| / \text{accepted} \times 100\%$$

accepted value: $g = 9.81$ m/s²

Then apply the rule from the lab-writing guidelines. **Under 10%**, you may attribute the difference to ordinary experimental uncertainty and say so. **Over 10%**, that is no longer an acceptable explanation: you must name a specific physical mechanism and *put a number on it*. Expect to be in the second case. The tape method has a real, identifiable systematic error in it, and finding it is the point of the lab.

OPTIONAL **A second route to the same number — up to 3 bonus points**

You do not need this section to finish the lab or to earn full marks. Everything above is the complete assignment. What follows is a second, independent way to get g out of the same tape, for anyone who wants to try it — it is worth up to three bonus points if you do it correctly and write it up properly.

The fit in Part III found the acceleration from the *curvature* of a parabola. You can also get it from the *slope of a straight line*, by turning your positions into velocities first.

The dot spacing is already velocity information: over one tick the mass moves some distance, so dividing that distance by the tick time gives a velocity. The cleanest version uses the dots on *either side* of the one you are interested in:

$$v_n = (x_{n+1} - x_{n-1}) / (2\Delta t)$$

central-difference velocity at dot n, where $\Delta t = 1/f$

This is the average velocity across the two ticks straddling dot n, and because the motion has constant acceleration, that average is exactly the instantaneous velocity *at* dot n — not an approximation. You can compute it for every dot except the first and the last. Then, from Module 1, $v = v_0 + a\Delta t$, which is a straight line.

If you want to try it

1. Add a velocity column next to one tape's data: `=(x_next − x_prev)/(2*Δt)`, in m/s. Leave the first and last rows blank — those dots have no neighbour on one side.
2. Plot velocity against time as a Scatter (XY) chart.
3. Add a trendline — this time **Linear** — with the equation and R^2 displayed.
4. The **slope is your acceleration directly**, in m/s². No doubling. The intercept is v_0, and it should come out close to the B from your order-2 fit.

To earn the bonus, include the velocity-versus-time graph as a numbered figure with its fit equation and R^2, show one sample calculation of a central-difference velocity with units, and add two or three sentences to your discussion comparing the two accelerations — how close are they, and what does it mean that two different reductions of the same measurements agree or disagree?

One tape is enough for the bonus. You do not have to do all three.

Check your fit

Paste one tape's distances below — one per line, starting with dot 0 — and this tool will fit them and report the numbers you should be seeing in your spreadsheet. Use it to catch arithmetic and units mistakes. **It is a check on your own work, not a**

substitute for it: your report must show the spreadsheet chart, the trendline equation, and a sample calculation you did by hand.

Σ TOOL · ORDER-2 FIT TO ONE TAPE

Timer frequency (Hz)

40

Units of the values you paste

centimetres (cm)

Distance from dot 0 — one value per line, dot 0 first

Percent error is computed against $g = 9.81\ \text{m/s}^2$. The last two chips show the optional velocity route, for anyone attempting the bonus — ignore them otherwise.

Questions for your discussion

Work these into your **Discussion** and **Additional Questions** sections, in sentences and paragraphs — never as a numbered list, and never by restating the question before answering it. Do not limit yourself to these; add any other insight you have.

I. Your fit reported a value for B. Convert it into a time: if the mass had been falling freely from rest, how long had it been falling when it reached dot 0? Compare that to one tick of the timer. Does the number you get match the size of the blot you skipped past at the start of the tape?

II. State which direction your measured g missed 9.81, and propose a physical mechanism that is *consistent with that direction*. A mechanism that would push the answer the wrong way is evidence against itself.

III. Treat the tape's drag as a constant retarding force and estimate its size per unit mass from your own deficit: if you measured g_{meas}, then the drag contributes $(9.81 - g_{\text{meas}})$ in m/s^2. Multiply by your hanging mass to get a force in newtons. Is that a plausible amount of friction for paper sliding past a vibrating stylus? Say why or why not.

IV. Your R^2 values will be very close to 1. Say precisely what that does and does not tell you. Can a data set be shifted by a systematic error and still produce $R^2 = 0.9999$?

V. The time axis of this experiment carries almost no uncertainty, while the distance axis carries all of it. Explain why, and say which of the two is limiting how well you know your answer.

VI. Suppose your three tapes agree with each other to within $0.1\ \text{m/s}^2$ while sitting a full $1\ \text{m/s}^2$ away from 9.81. Explain in your own words why that combination is *more* troubling than a large scatter would be, not less.

VII. Name one change to the apparatus or the procedure that would reduce the dominant error here, and say roughly how much of the deficit you would expect it to remove.

> **One thing the rubric will not accept:** "human error." It explains nothing, and it is not a category of error in experimental physics. If your percent error is under 10%, attribute it to ordinary experimental uncertainty and say so. If it is over 10%, you must identify specific error sources and *quantify* them — a named mechanism with a number attached.

✔ Before you submit

Every line here maps to a row on the Lab 02 grading rubric.

FORMAT & FRONT MATTER

- [ ] Lab Template used, with names, course, lab number, and both dates in the heading
- [ ] Abstract: four sentences or fewer — theme, numerical results for g, percent error. Nothing else.
- [ ] Introduction states the kinematic equation and the parabola it maps onto, defines every variable with units, and ends with the question the lab answers

EXPERIMENTAL

- [ ] Timer frequency stated explicitly, read off the unit you used
- [ ] Equipment described within the procedure, in paragraphs — not as a bare list
- [ ] Procedure in your own words, detailed enough to follow blindly — not copied from this handout
- [ ] A photo of the setup plus a labelled sketch showing where dot 0 is and which direction was measured as positive
- [ ] How you chose dot 0, and how many dots you measured on each tape

RESULTS

- [ ] Raw dot data table for all three tapes — numbered, captioned, units on every column, referenced in the text
- [ ] Fit-results table with A, B, C, R^2, $g = 2A$ and percent error
- [ ] Position-vs-time graph for each tape, with the order-2 trendline, its equation and R^2 displayed
- [ ] Every graph numbered as a Figure, captioned, and described in the text ("Figure 2 shows...")
- [ ] A sample calculation, worked step by step with units, for *each* type: cm→m conversion, dot number to time, $a = 2A$, mean, standard deviation, percent error
- [ ] Significant digits handled correctly throughout, including in the tables

DISCUSSION & CONCLUSION

- [ ] Opens with "we experimentally determined that… ± error"
- [ ] Result compared to 9.81 m/s² by percent error
- [ ] The 10% rule applied — and if over 10%, a named mechanism with a number attached
- [ ] *B* interpreted physically, not just reported
- [ ] The *N* = 3 limitation acknowledged
- [ ] Additional questions answered in a dedicated section, in paragraphs
- [ ] The phrase "human error" appears nowhere in the report
- [ ] Conclusion: 3–4 sentences — numerical result with percent error, a definitive statement, and one concrete improvement

REFERENCES

- [ ] At least one source beyond the textbook and this handout — on *g* and how it varies with latitude and altitude
- [ ] Author, title and date given for each — a bare URL is not a citation

OPTIONAL — BONUS ONLY

- [ ] Velocity-vs-time graph for at least one tape, numbered and captioned, with its linear fit equation and R^2
- [ ] Sample calculation of one central-difference velocity, with units
- [ ] Two or three sentences comparing the two accelerations and saying what the agreement or disagreement means

0 of 27 checked.

Files for this lab

HANDOUT · PDF

This lab, printable

DATA SHEET · XLSX

Raw data sheet

The whole page as a PDF. Print it and bring it to lab — the interactive checker only works on this page, but everything else is there.

Type your dot distances into the shaded cells; time, position and the summary calculate themselves. Set the timer frequency at the top first.

GRADING · DOCX

Lab 02 rubric

Exactly what your report is marked on, criterion by criterion, with the full-credit description for each. Read it before you write, not after — it doubles as an outline.

WRITING AID · DOCX

Example starter sentences

A model report with XX placeholders where your numbers go, and a prompt for every table and figure the rubric asks for. Not a report to hand in — a way to start one.

REQUIRED · ON BRIGHTSPACE

Lab Template and Writing Lab Reports module

Not linked here — find them in the **Writing Lab Reports** module on Brightspace. Read both before you write: formatting, the abstract rules, significant digits, and what belongs in each section.

References to get you started

Your references should include material on *g* and on free fall. Your textbook, this handout, and the Module 1 note pages all count, and you need at least one source beyond the textbook. Use this format:

[1] [Authors' names], *Journal or Book title*, edition, Volume number, page number, (year).

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Revised from the PHY 151 Free Fall lab by H. Trevor Johnson-Steigelman. Ticker-tape portion retained; the remote-scan and sonic-ranger sections have been retired in favour of data the class collects itself. Revised 2026-09-21.